



SAFETY DATA SHEET

REFINED GLYCERINE

Section 1 - Chemical Product Identification

MSDS Name: Refined Glycerine

Synonyms: Glycerol, 1,2,3-Propanetriol, 1,2,3-Trihydroxypropane

CAS No.: 56-81-5

Chemical Formula: $C_3H_8O_3$

Product Use: Pharmaceutical, food, cosmetics and industrial applications

Company name: Thai Glycerine 102,102/3 Moo 3 Tambol Thasao, Amphur Krathumbaen, Samutsakorn 74110

Section 2 - Hazards Identification

GHS Classification of Substance in accordance to Regulation (EC) No. 1272/2008 (CLP / GHS) : Not a hazardous substance or mixture .

OSHA Hazard Communication Standard This product is not a " Hazardous Chemical " as defined by the OSHA Hazard Communication Standard , 29 CFR 1910.1200 .

GHS Label Elements : None

Other Hazards

Potential Health Effects :

Can be irritating to the eyes .

Can be harmful if ingested .

Can be harmful if inhaled . Avoid breathing mist .

Can be irritating to the skin .

Section 3 - Composition, Information on Ingredients

Chemical Name	CAS Number	EU Number	Percent	EU Classification	GHS Classification
Refined Glycerine	56-81-5	200-289-5	99.5% Min	Not Classified	Not Classified

Section 4 - First Aid Measures

Description of first aid measures

If inhaled : Move to fresh air.

In case of skin contact : Wash off with soap and water.

In case of eye contact : Rinse thoroughly with plenty of water, also under the eyelids. If eye irritation persists, consult a specialist.

If swallowed : Clean mouth with water and drink afterwards plenty of water.

Never give anything by mouth to an unconscious person. If symptoms persist, call a physician.

Most important symptoms and effects, both acute and delayed

Symptoms : See Section 11 for more detailed information on health effects and symptoms.

Effects : See Section 11 for more detailed information on health effects and symptoms.

Indication of any immediate medical attention and special treatment needed

Treatment : Treat symptomatically.

Section 5 - Fire Fighting Measures

Suitable (and unsuitable) extinguishing media

Suitable extinguishing media: Powder, alcohol-resistant foam, water spray, carbon dioxide, sand

Unsuitable extinguishing media: No data available.

Specific hazards arising from the chemical: No data available.

Special protective equipment and precautions for firefighters

Special fire fighting procedures: Keep away from sources of ignition and use appropriate extinguishing media.

Fight fire from upwind position if possible.

Special protective equipment for fire-fighters: Use goggles in combination with dust mask, and other protections as appropriate to situation.

Section 6 - Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Advice for non-emergency personnel: Do not breathe vapors, aerosols. Evacuate the danger area, observe emergency procedures, consult an expert.

For personal protection see section 8.

Environmental precautions

Do not let product enter drains.

Methods and materials for containment and cleaning up

Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions (see sections 7 and 10). Take up with liquid-absorbent material (e.g. Chemisorb®).

Dispose of properly. Clean up affected area.

Section 7 - Handling and Storage

Handling: No special precautions required, but avoid eye and skin contact as part of normal industrial hygiene. Prevent formation of mist. Eye and skin contact should be avoided if handling at elevated temperatures.

Storage: Keep in a tightly closed container and store upright to prevent any run-out of product. Store in a cool, dry, well ventilated area, away from incompatible substances.

Refer to section 6 for clean-up of spillages.

Refer to Section 13 for disposal considerations.

Section 8 - Exposure Controls, Personal Protection

Engineering Controls: No special measures required.

Ventilation: Local exhaust preferred. Mechanical (general) acceptable.

Personal Protective Equipment:

Respiratory Protection: If the exposure limit is exceeded and engineering controls are not feasible, a half face piece particulate respirator (NIOSH type P95 or R95 filters) may be worn for up to ten times the exposure limit, or respirator supplier, whichever is lowest. Please note that N filters are not recommended for this material.

Hand Protection: Suitable protective gloves.

Body Protection: Chemical boots and full protective clothing may be required.

Eye Protection: Protective goggles or face shield with goggles.

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Section 9 - Physical and Chemical Properties

Physical state : Viscous Liquid	Decomposition temperature : 290 °C
Molecular mass : 92.09 g/mol	Flammability (solid, gas) : No data available
Colour : Colourless.	Vapour pressure : No data available
Odour : characteristic.	Vapour pressure at 50 °C : 0.0025 mbar
Odour threshold : No data available	Relative vapour density at 20 °C : 3.18
pH : Neutral to litmus	Relative density : No data available
Relative evaporation rate (butylacetate=1) : No data available	Density : 1.2613 g/ml
Vapor Density: 3.17 (Air=1)	Specific Gravity : 1.26 at 25°C
Evaporation Rate: Not available	Solubility : Soluble in water. Soluble in ethanol.
Viscosity: 525 mPa.s at 25°C	Log Pow : 0
Melting point : Approx. 18°C	Log Kow : No data available
Freezing point: No data available	Viscosity, kinematic : 1181.3 cSt
Boiling point : Approx. 182 °C	Viscosity, dynamic : No data available
Flash pointn : Approx. 180 °C	Explosive properties : No data available
Self ignition temperature : 400 °C	Oxidising properties : No data available
	Explosive limits : ≥ 1.1 vol %

Section 10 - Stability and Reactivity

Reactivity: No relevant information available.

Chemical stability: Stable under normal temperatures and pressures.

Thermal decomposition / conditions to be avoided: No decomposition if used and stored according to specifications.

Possibility of hazardous reactions

Reacts with strong acids and alkali.

Reacts with strong oxidizing agents.

Toxic fumes may be released if heated above the decomposition point.

Conditions to avoid Excessive heat.

Incompatible materials Oxidizers, strong bases, strong acids

Hazardous decomposition products Does not decompose up to 204 °C (400 °F) Thermal decomposition may release acrolein.

Section 11 - Toxicological Information

Information on hazard classes as defined in Regulation (EC) No 1272/2008 Test data are not available for the complete mixture.

Classification procedure The method for classification of the mixture is based on ingredients of the mixture (additivity formula).

Classification according to GHS (1272/2008/EC, CLP) This mixture does not meet the criteria for classification in accordance with Regulation No 1272/2008/EC.

Acute toxicity Shall not be classified as acutely toxic.

Skin corrosion/irritation Shall not be classified as corrosive/irritant to skin.

Serious eye damage/eye irritation Shall not be classified as seriously damaging to the eye or eye irritant.

Respiratory or skin sensitization Shall not be classified as a respiratory or skin sensitizer.

Germ cell mutagenicity Shall not be classified as germ cell mutagenic.

Carcinogenicity Shall not be classified as carcinogenic.

Reproductive toxicity Shall not be classified as a reproductive toxicant.

Specific target organ toxicity - single exposure Shall not be classified as a specific target organ toxicant (single exposure).

Specific target organ toxicity - repeated exposure Shall not be classified as a specific target organ toxicant (repeated exposure).

Aspiration hazard

Shall not be classified as presenting an aspiration hazard.

Symptoms related to the physical, chemical and toxicological characteristics

- **If swallowed** diarrhoea, vomiting, gastrointestinal complaints
- **If in eyes** slightly irritant but not relevant for classification
- **If inhaled** dizziness, headache, slightly irritant but not relevant for classification
- **If on skin** Frequently or prolonged contact with skin may cause dermal irritation

Section 12 - Ecological Information

Toxicity Shall not be classified as hazardous to the aquatic environment.

Biodegradation Data are not available.

Process of degradability Data are not available.

Bioaccumulative potential Data are not available.

Mobility in soil Data are not available.



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Results of PBT and vPvB assessment Data are not available.

Endocrine disrupting properties None of the ingredients are listed.

Other adverse effects Data are not available.

Section 13 - Disposal Considerations

Waste treatment methods

· Recommendation:

Smaller quantities can be disposed of with household waste.

The user of this material has the responsibility to dispose of unused material, residues and containers in compliance with all relevant local, state and federal laws and regulations regarding treatment, storage and disposal for hazardous and nonhazardous wastes.

· Uncleaned packagings

· Recommendation: Disposal must be made according to official regulations.

· Recommended cleansing agent: Water, if necessary with cleansing agents.

Section 14 - Transport Information

UN number or ID number not subject to transport regulations

UN proper shipping name not assigned

Transport hazard class(es) none

Packing group not assigned

Environmental hazards non-environmentally hazardous acc. to the dangerous goods regulations

Special precautions for user There is no additional information.

Maritime transport in bulk according to IMO instruments The cargo is not intended to be carried in bulk.

Information for each of the UN Model Regulations

Transport of dangerous goods by road, rail and inland waterway (ADR/RID/ADN) – Additional information

Not subject to ADR, RID and ADN.

International Maritime Dangerous Goods Code (IMDG) - Additional information Not subject to IMDG.

International Civil Aviation Organization (ICAO-IATA/DGR) - Additional information Not subject to ICAO-IATA.

Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

This material safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.

National legislation

Seveso III: Directive 2012/18/EU of the European Parliament and of the Council on the control of major-accident hazards involving dangerous substances.

Not applicable

Section 16 - Other Information

The information presented herein is believed to be factual as it has been derived from the works and opinions of persons believed to be qualified experts; however, nothing contained in this information is to be taken as a warranty or representation for which Thai Glycerine bears responsibility. The user should review any recommendations in the specific context of the intended use to determine whether they are appropriate